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PROJECT

Network Segmentation Through VLAN Implementation

CYB320, Intermediate Routing and Switching

Network Security

VLANs

Routing

Research

Technical Writing

Overview

This project examines how VLAN segmentation and inter-VLAN routing can improve security, reduce broadcast traffic, and support scalable enterprise network design. It was completed as the capstone research assignment for CYB320.

Network Diagram

Network topology diagram: three VLANs connected through a Layer 3 switch

Approach

I researched three enterprise network scenarios and compared flat network designs against segmented designs using VLANs for departmental separation. I modeled traffic flow using Packet Tracer and documented how inter-VLAN routing rules affect both performance and access control.

Challenge

The most difficult part was balancing security segmentation with usability. Overly strict routing rules blocked traffic that departments legitimately needed to share, which required revisiting the access control list design twice before it worked as intended.

Result

The final design reduced unnecessary broadcast traffic by isolating departments into their own VLANs while preserving controlled access between finance and IT systems. The project received a grade of 96 percent and was highlighted by my instructor as a strong example of applied network design reasoning.

Reflection: This project strengthened my understanding of how design decisions affect both security and performance. If I revisited this project today, I would add a formal risk assessment comparing the segmented design against a zero-trust alternative.

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